

Tien Soare (Nguyen)

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OBJECTIVE

Ph.D. researcher in Computer Science investigating how large language models can advance software engineering practices in testing, debugging, and software reliability. Designs and evaluates repair and verification techniques across large corpora of real-world scientific and data-centric code.

TECHNICAL SKILLS

Languages: Python, Java, SQL, JavaScript, HTML/CSS

ML & LLM Systems: vLLM (batch inference serving), quantization (FP8/INT8 evaluation), model evaluation and benchmarking, prompt design, PyTorch, HuggingFace Transformers

HPC & Infrastructure: Slurm, ARC/TinkerCliffs (NVIDIA A100), Docker, Conda, joblib, diskcache

Program Analysis: AST analysis, symbolic computation (SymPy), numerical stability, floating-point verification, pytest, JUnit

Developer Tools: Git, VS Code, PyCharm, Jupyter, Papermill, Cursor, LaTeX

Web: React, Flask, FastAPI, REST APIs, Material-UI

WORK EXPERIENCE

Graduate Research Assistant

August 2023 – Present

Virginia Tech

Blacksburg, VA

- Built and ran large-scale LLM evaluation pipelines on Virginia Tech's ARC cluster, orchestrating vLLM inference across A100 GPUs under Slurm for 469K prompts and 3B tokens
- Developed static and dynamic analysis tooling for Python programs, including AST-based semantic-equivalence checking and symbolic verification with SymPy
- Collaborated with two faculty advisors and a research group across four concurrent projects, with regular written progress reporting

Information Security Student Coordinator

Feb. 2023 – May. 2023

University of Saint Thomas

St. Paul, MN

- Monitored and secured university systems and networks within the Security Operations Center (SOC)
- Conducted routine security audits to address about 20 system incidents per day
- Resolved about 5 IT service tickets per day using management tools for optimal efficiency
- Managed Azure Active Directory users, adhering to FERPA guidelines

Software Engineering Intern

Jun. 2021 – Aug. 2021

Target

Minneapolis, MN

- Built a full-stack web application that consolidated prepaid product masking codes from scattered sources into a single searchable tool, giving 100+ authorized users one reliable lookup point
- Developed responsive front-end components with React, JavaScript, Material UI, and Axios, streamlining search and retrieval for end users
- Implemented REST API endpoints on the back end using Java, Spring Boot, and Oracle Database to serve masking-code queries at scale
- Collaborated in an Agile team, using Jira for weekly sprint planning to deliver features iteratively

PUBLICATION

Are the Majority of Public Computational Notebooks Pathologically Non-Executable?

Tien Nguyen, Waris Gill, and Muhammad Ali Gulzar. Accepted to MSR 2025. EMSE Special Invitee.

- Analyzed 42,546 public Jupyter notebooks using static and dynamic approaches, introducing the first formal definitions of partial and pathological non-executability
- Classified notebooks into fully executable, partially executable, restorable, or pathologically non-executable categories based on runtime and dependency failures
- Found that only 21.3% of notebooks are truly pathologically non-executable, much lower than the previously reported estimation of up to 76%

- Applied LLM-based generative techniques for code synthesis and repair, framing the task as structured generation and transformation, achieving full restoration for 5.4% and partial restoration for 12.8% of non-executable notebooks

Investigating and Remediating Pathological Non-Executability of Public Computational Notebooks

Tien Nguyen, Waris Gill, and Muhammad Ali Gulzar. *Extension of MSR 2025.*

- Built an automated testing tool to evaluate each notebook's executability, checking for required modules, external input files, and linear execution order
- Evaluated a curated dataset of 3,169 public notebooks, generating over 17,000 test cases and observing an average pass rate of 73.1%

Assessing Large Language Models for Stabilizing Numerical Expressions in Scientific Software

Tien Nguyen, Kirshanthan Sundararajah, and Muhammad Ali Gulzar. *In submission to ICSE 2027.*

- Designed the first evaluation of LLM reasoning for high-precision numerical computation across two tasks: *detection* (generating error-inducing inputs to expose instability in numerical expressions) and *stabilization* (rewriting expressions for numerical stability), verified by a two-stage semantic-equivalence check (structural AST analysis + numerical validation) to guarantee rewrites preserve semantics
- Curated a benchmark of 2,037 expressions and evaluated 4 LLMs over ~469K prompts and 3B tokens
- Found that while classical baselines stabilize more expressions overall (93.7% vs. 67.2%), LLMs uniquely fix 61.2% (79/129) of cases where baselines fail and beat baseline stability on 24% (489) of all expressions, positioning LLMs as a complementary tool for numerical robustness

EDUCATION

Virginia Tech	Blacksburg, VA
<i>Ph.D. in Computer Science</i>	<i>Anticipated graduation: May 2028</i>
Virginia Tech	Blacksburg, VA
<i>Master's in Computer Science</i>	<i>Aug. 2023 – May 2026</i>
University of Saint Thomas	St. Paul, MN
<i>Bachelor in Computer Science, minor in Theology</i>	<i>Aug. 2020 – May 2022</i>
Century College	White Bear Lake, MN
<i>Associate's in Computer Science</i>	<i>Jan. 2018 – May 2020</i>

PROJECTS

Bookmark Management | *Virginia Tech* | [github](#)

- Developed interactive frontend using React.js and CSS for efficient bookmark management in a full-stack web app
- Collaborated with the team on backend integration (Node.js, DBeaver, and PostgreSQL) and deployed the app on a local server

Book Library | *Personal Project* | [github](#)

- Built a React web app implementing search and ranking functionality over book metadata using the Google Books API, including data cleaning and error handling
- Implemented pagination, sorting, and local storage for favorite books, managing state with React hooks
- Designed responsive UI with Bootstrap, styled-components, and integrated icons via react-icons

Classy Schedule | *University of St. Thomas* | [github](#)

- Developed a web application allowing department chairs to schedule classes and teaching sections
- Designed and implemented front-end UI components using React, Material UI, and JavaScript
- Worked with backend and database teams to integrate and deploy the app on Azure

AbroadstudiZ | *gAlpha, University of St. Thomas* | [youtube](#)

- Designed a web platform for international students using bubble.io, enabling study abroad planning and experience sharing
- Interviewed 23 international students and 6 international officers to identify pain points and refine the platform
- Selected as one of 6 finalists in the 2021 Fowler Global Social Innovation Challenge